

```

Administrator: Command Prompt - python update4.py

:\sherin_predict>python update4.py
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
0000 00:00:1776922210.942225    2288 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to float
rn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
0000 00:00:1776922218.032411    2288 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to float
rn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
firebase Connected
system Started
VERLOAD: 1800.0 mW
aved: 2026-04-23 11:00:22 1800.0 mW
VERLOAD: 1800.0 mW
aved: 2026-04-23 11:00:27 1800.0 mW
VERLOAD: 1800.0 mW
aved: 2026-04-23 11:00:33 1800.0 mW
VERLOAD: 1800.0 mW
aved: 2026-04-23 11:00:39 1800.0 mW
aved: 2026-04-23 11:00:44 0.0 mW
aved: 2026-04-23 11:00:50 0.0 mW
aved: 2026-04-23 11:00:56 0.0 mW
aved: 2026-04-23 11:01:01 0.0 mW
VERLOAD: 2300.0 mW
aved: 2026-04-23 11:01:07 2300.0 mW
VERLOAD: 2100.0 mW
aved: 2026-04-23 11:01:13 2100.0 mW
VERLOAD: 2100.0 mW
aved: 2026-04-23 11:01:18 2100.0 mW

```

Fig. 6: Python console output

```

===== PZEM DATA =====
Voltage   : 29.80 V
Current   : 60.07 mA
Power     : 1790.00 mW
Energy    : 0.00 kWh
Frequency : 50.00 Hz
PF        : 1.00
⬆️ Uploaded to Firebase
Fri Jul 05 2019 12:21:16

WiFi: CONNECTED | IP: 192.168.1.101
Configuring Firebase...
Firebase URL: https://smartgrid-monitoring-default-rtdb.firebaseio.com/
Database Path: /SmartGrid/Live
User Email: smartgrid@gmail.com
User UID: 5eJxj3abcDEF123456
Project ID: smartgrid-monitoring
Database Secret: *****
Firebase Ready
Waiting for PZEM data...
✔️ Smart Grid Monitoring Started (24V Model)

===== PZEM DATA =====
Voltage   : 29.80 V
Current   : 60.07 mA
Power     : 1790.00 mW
Energy    : 0.00 kWh
Frequency : 50.00 Hz
PF        : 1.00
⬆️ Uploaded to Firebase

===== PZEM DATA =====
Voltage   : 29.80 V
Current   : 60.07 mA
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===== PZEM DATA =====
Voltage   : 29.80 V
Current   : 60.07 mA
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Frequency : 50.00 Hz
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⬆️ Uploaded to Firebase

```

Fig. 7: The serial monitor output

while the ferrite-core current transformer (CT) measures the load current by sensing the current in the live wire. The PZEM-004T communicates with the ESP32 through its UART interface (TX/RX pins), while the 5V and GND pins provide power to the module.

When AC power is applied, the PZEM-004T continuously measures line voltage, load current, active power, energy consumption, frequency, and power factor. The ESP32 periodically reads these parameters, processes the data, and uploads it to the Firebase real-time database over Wi-Fi for remote monitoring.

A Python-based AI application retrieves historical data from Firebase, analyses load trends using machine learning, and predicts future load growth. The prediction results are stored in Firebase and can also be exported to an Excel file for reporting and further analysis. This enables continuous load monitoring, predictive analytics, and proactive transformer capacity planning.

EFY note. A 230V/12V-0-12V, 1A transformer is used for testing. In actual substations, high-voltage power transformers (for example,